Forsíða Viðmótsforritun Lokapróf 2025

Midterm Exam HBV201G Viðmótsforritun

Teacher: Ebba Þóra Hvannberg (fyrirlestrar)

Faculty: Industrial engineering, Mechanical Engineering and Computer Science

Course: HBV201G Viðmótsforritun

ETCS: 8

Weight of exam: 49%-69% of final grade

When: [9:00-12:00]

- **Exam rules:** The exam is taken in the Inspera testing system on students' computers that are configured to be locked with Safe Exam Browser (SEB).
- **Supporting material:** No supporting material allowed
- **Resources:** Pay special attention to the resources that come with the exam.
- **Structure of the exam** includes multiple choice questions. There are several answer options listed and the student chooses the "correct or most correct" answer option each time. If a student is to select more than one option it is stated. There is no deduction for incorrect answers.
- **Taking the exam:** It is important to read the question carefully before selecting a final answer. After answering a question, press the "Next" button or select the next question on the numbered flow line at the bottom of the screen. This action automatically saves the answer, but the student's work is also automatically saved every 15-20 seconds. It is possible to go back and forth in the test and mark questions that remain to be answered.
- **Assessment:** The exam is graded is automatically calculated based on multiple choice questions. Java questions in an open editor are graded according to a rubric.

Good luck

Ebba Þóra Hvannberg

HBV201G Lokapróf 2025 Inngangur að hluta 1

Part I. Multiple-choice questions. Total 45%. Open-ended questions in Part II count 55%.

In the multiple-choice questions, only one answer is possible. There is no deduction for an incorrect answer.

If you cannot see part of the exam, e.g. pictures or supporting documents, please contact the invigilator.

If a defect in the exam is discovered, it will be reviewed during the exam. If a student is unsure whether his understanding of a question is correct, he can indicate what that understanding is in the answer box for open-ended questions and/or send an email to the teacher after the exam.

The following supporting documents are included

1 https://prof.snara.is/

2 https://www.bing.com/translator

3 https://openjfx.io/javadoc/23/

4 https://openjfx.io/javadoc/23/javafx.graphics/javafx/scene/doc-files/cssref.html

5 Java cheatsheet

6 Additional cheatsheet for programming questions

7 currency-view.fxml

8 media-view.fxml

1 JavaFX og SceneBuilder 1.1

Which of the following statements correctly describes how a JavaFX application is set up using the `Application` class, an FXML file, and a connection to a `Controller`?

Select one alternative:

- The Application class loads the interface from a Java class file using the fx:id setting.
- To work with .fxml files in IntelliJ, it is not necessary to specifically link SceneBuilder to IntelliJ.
- JavaFX runs directly on the backend application without requiring the Java SDK.
- SceneBuilder saves the user interface as Java code, which is loaded directly into the Application class.
- A JavaFX application consists of an Application class that launches an .fxml file and a Controller that connects the user interface to events. ✔

Maximum marks: 3

2 Region klasinn - 3.3

In JavaFX, the `Region` class is used to set the visual properties of a UI component. Which of the following methods correctly shows how to adjust the shape of a UI component so that it appears as a circle?

Select one alternative:

- region.setShape(new Shape(50));
- region.setClip(new Circle(50));
- region.setNodeShape(Circle.class);
- region.setStyle("-fx-shape: circle;");
- region.setShape(new Circle(50)); ✔

Maximum marks: 3

3 Properties og binding - 4.1

In JavaFX, it is possible to bind properties together so that changes in one affect another. Given two `TextField` objects, `textField1` and `textField2`, which of the following methods shows how to bind `textField2` so that it always displays the text from `textField1`?

Select one alternative:

- textField2.textProperty().bindBidirectional(textField1.getText());
- textField2.textProperty().set(textField1.textProperty());
- textField2.setText(textField1.getText());
- textField2.bind(textField1.textProperty());
- textField2.textProperty().bind(textField1.textProperty()); ✔

Maximum marks: 3

4 Menus - 7.1

In JavaFX programming, it is common to use menus to allow the user to control the interface. Which of the following statements about `Menu` and `MenuItem` is correct?

Select one alternative:

- A MenuBar contains Menu objects, and each Menu contains MenuItem objects. ✔
- A MenuItem cannot have an event handler attached to it.
- A Menu always contains only one MenuItem.
- A ContextMenu only works when the right mouse button is used.
- A MenuBar object cannot be created using FXML.

Maximum marks: 3

5 Hönnun notendaviðmóts - 3.1

You are designing a user interface where the user selects a date and clicks a button to confirm the selection. Which of the following implementations best follows recognized design guidelines, particularly the principles of action proximity and object clarity?

Select one alternative:

- The date picker is in a popup window, and the button is on a toolbar.
- The button is on the left side and the date picker is on the right side, each in opposite corners.
- The date picker and the button are side by side, centered in the window. ✔
- The button is at the top and the date picker is at the bottom of the window.
- All UI components are placed together in one corner of the screen to optimize space usage.

Maximum marks: 3

6 Teikning og changeListener - 8.1

Here is a snippet from a JavaFX program where two rectangles (`r` and `fxSvartur`) are displayed on the screen. The variable `r` has a `ChangeListener` attached to its `xProperty`:

```
r.xProperty().addListener((obs, oldVal, newVal) -> {
    if (r.getBoundsInParent().intersects(fxSvartur.getBoundsInParent())) {
        fxSvartur.setFill(Color.MAGENTA);
    } else {
        fxSvartur.setFill(Color.BLACK);
    }
});
```

What is the purpose of this part of the code?

Select one alternative:

- The program checks whether r and fxSvartur have the same parent and changes the color if they do.
- fxSvartur automatically moves to the same x-coordinate as r.
- A mouse click on r triggers this response and adds a new rectangle.
- r resets to its initial position if it overlaps with fxSvartur.
- The program checks whether r overlaps with fxSvartur and changes the color of fxSvartur to pink if it does. ✔

Maximum marks: 3

7 Viðtöl við væntanlega notendur hugbúnaðar 9.4

You are about to interview a potential user of the software you are designing. You want to elicit useful answers and build a trusting relationship with the interviewee. Which of the following approaches to questioning and interview behavior best supports this goal?

Select one alternative:

- I avoid leading questions and listen without judgment, asking for real examples to elicit honest answers. ✔
- I create prepared answer options to promote better flow in the interview and make it easier for the interviewee to respond.
- I start the interview by explaining my ideas and hope that the interviewee will support them.
- I try to move quickly through the list of questions so that the interview does not become too long.
- I ask general questions and let the interviewee talk about whatever comes to mind, without narrowing down the discussion.

Maximum marks: 3

8 Dialog og resultConverter í JavaFX 5.2

Look at the following snippet from a JavaFX dialog class:

```
public class TrainerDialog extends Dialog<Trainer> {
    public TrainerDialog() {
        DialogPane pane = FXMLLoader.load(getClass()
              .getResource("trainer-view.fxml"));
        setDialogPane(pane);
        setResultConverter(button -> {
            if (button == ButtonType.OK) {
                Trainer t = new Trainer();
                t.setNafn(fxNafn.getText());
                return t;
            }
            return null;
        });
    }
}
```

Which of the following describes how data is sent from the dialog?

Select one alternative:

- When the user clicks the OK button, the resultConverter retrieves data from the interface and returns a Trainer c ✔ ct.
- showAndWait() only returns values when a TextInputDialog is used, but not with custom dialogs like Dialog<Trainer>.
- FXML describes the DialogPane, but the resultConverter that returns a Trainer is implemented in code.
- setResultConverter is only used with Alert<Trainer> for this purpose.
- JavaFX runs the resultConverter in the background without returning a Trainer object.

Maximum marks: 3

9 CSS og styleClass í JavaFX 3.2

```
.input-green {
-fx-border-color: green
}
.input-red {
-fx-border-color: red
}

<TextField fx:id="inputField" styleClass="input-green" />

inputField.getStyleClass().removeAll("input-green");
inputField.getStyleClass().add("input-red");
```

Look at the snippets from `.css`, `.fxml`, and Java code that modifies the appearance of a `TextField` UI component in a JavaFX application:

Which statement correctly describes how JavaFX works with `styleClass` and styling in this context?

Select one alternative:

- When two CSS style rules equally apply to the same UI component, the one defined later in the stylesheet determines its appearance. ✔
- styleClass can only be used in Java code, not in .fxml.
- Using clear() only removes custom styles, not the default styleClass.
- It is not possible to link a stylesheet to a TextField directly from the .fxml file.
- getStyleClass() automatically removes the old style if a new one is added with add()

Maximum marks: 3

10 Maven og pom.xml - 1.2

You are working on a Java project that uses Maven. Which of the following statements about the `pom.xml` file is correct?

Select one alternative:

- The pom.xml is only used to set fonts (styles) in SceneBuilder.
- The pom.xml defines which libraries (dependencies) are needed, the Java version, and the build and runtime configurations. ✔
- The pom.xml is only used in Java projects.
- The pom.xml file must be located in the src/main/java folder.
- The pom.xml is a code file that needs to be compiled separately.

Maximum marks: 3

11 Skipting milli viðmóta í sama glugga með ViewSwitcher 5.1

```
ViewSwitcher.setScene(scene);
ViewSwitcher.switchTo(View.LOGIN);
```

In the following code, the interface (.fxml) is being switched within the same window in a JavaFX application using `ViewSwitcher` and the `View` enum:

Which of the following statements correctly describes how this method works?

Select one alternative:

- ViewSwitcher.switchTo(View.LOGIN) sets the root node in the same scene and keeps the window open. ✔
- The switchTo() method closes the old scene and creates a new window for each scene.
- The View items are in an enum where each one stores a complete Scene and Stage
- switchTo() can only use interfaces that have not been previously displayed in the application.
- It is possible to use ViewSwitcher.switchTo() without first calling setScene(scene).

Maximum marks: 3

12 Hópvinna og góðar Git venjur

You are in a three-person team working on a programming project that is due at the end of the semester. Which of the following approaches to project management and communication best supports both the team's success and the visible contribution of individual members, in line with good practices?

Select one alternative:

- The project is hosted on GitHub, and the team works on the same branch with regular commits, pushes, and coordination to avoid conflicts. ✔
- Each member works independently on their own version of the code and merges it at the end by sending a single email.
- The team uses a shared Google Doc for coding and collaboration but does not store the code elsewhere.
- The team meets as needed and communicates via a messenger group when someone encounters problems.
- The team leader submits everything and lists all members as co-authors.

Maximum marks: 3

13 Git commit og lýsingar 11.2

```
@FXML
private void handleSendaNafn() {
    String nafn = fxTextField.getText();
    if (!nafn.isEmpty()) {
        fxSvar.setText("Velkomin/n, " + nafn + "!");
    } else {
        fxSvar.setText("Vinsamlegast sláðu inn nafn.");
    }
}
```

You have made the above additions to your project

Which description is most appropriate as Git commit messages for these additions?
Select one alternative:

- Added handling of the username in the interface, including conditional logic and response text. ✔
- Fixed a bug in the file and refactored some code.
- Dialog for players and CSS additions.
- Made minor changes to the text in the application.
- Pushed the latest changes.

Maximum marks: 3

14 Teiknimyndasögur og sviðssetning 10.2

You are tasked with creating a comic strip for an application that helps the user order food. Which of the following best describes how the setting and presentation of the comic strip should be?
Select one alternative:

- I show actions and the interface in frames with explanations below each frame, describing what is happening. ✔
- I sketch the idea and write the concept in text below the sketch.
- I draw the structure of the application and show all of its possibilities in each frame.
- I write a list of user steps but do not show screens or the interface.
- I use only text and omit drawings to keep the concept simple.

Maximum marks: 3

15 Val á frumgerð eftir verkefni 11.4

You are tasked with creating a prototype for a new web solution that is still in its early stages and aims to quickly test the core concept with potential users. Which type of prototype is most suitable?

Select one alternative:

- Paper prototype or sketches that show the user interface and flow without functional interactivity. ✔
- Service prototype with role-playing and scenario enactment in the reception.
- Hi-fidelity prototype built in a development environment with real functionality.
- A 3D model to assess the size and tactile properties of the interface.
- Wizard of Oz prototype where the system is manually controlled by a person.

Maximum marks: 3

ℹ HBV201G Lokapróf 2025 Inngangur að hluta 2

Part II

This part contains 4 questions worth a total of 55%.

For the programming qeustions, a Java editor is provided for programming in. The editor has statement numbers and displays Java highlights in color. The editor helps you with correct line indentation if you use curly braces. The editor does not check whether the program is grammatically correct and it does not translate or run the program.

Use best practices for programming. Import statements are not required

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16 Notendaprófanir - 12.1

Briefly describe how you would prepare, conduct, and analyze the results of user testing, with a focus on task selection, issue tracking, and severity assessment. Provide examples to support your explanation.

Fill in your answer here

Format

Words: 0

Maximum marks: 10

You are designing a page (scene) in a mobile app where the user can book an appointment with a doctor.
The user should be able to do the following actions

- Enter full name
- Select a doctor from a list
- Choose a date
- Choose a time
- Confirm the booking

The problem is in three parts. You will answer the first part on a sketch sheet and the next two parts in the editors with text and tables as described. Clearly label the parts.

1. Draw the scene using JavaFX UI components.
   Answer this part on a sketch sheet or, if you prefer, in the drawing section of the editor (pen).

2. Select UI components that are appropriate for each action (e.g., TextField, ComboBox, DatePicker, Button, or other).
   Set it up in a table (see the table icon in the editor) with two columns: Action; JavaFX UI Class.

3. Justify the design with reference to at least two items from the checklist below. Mention the items you used with the number and description.
   Set it up in a table (see the table icon in the editor) with two columns: Item and checklist number; How the item was used in the scene.

Checklist for User Interface Design

1. **Visibility** – Components should be visible and easy to find.
2. **Consistency** – UI elements should follow the same appearance and rules within the application.
3. **Familiarity** – Use words, symbols, and layouts that users recognize.

4. **Accessibility** – Ensure that text, colors, and buttons are accessible to all users.
5. **Provide information** – Users should know what is going on (feedback).
6. **Navigation** – It should be easy to move between steps or areas within the application.
7. **Control** – The user should be in control of their actions.
8. **Feedback** – The system should give feedback when an action is successful or fails.
9. **Error Recovery** – The user should be able to easily correct mistakes (e.g., Undo or rollback steps).
10. **Security** – Prevent the user from making critical errors.
11. **Flexibility** – Users should be able to adapt the appearance or options to their needs.
12. **Style** – The interface should look good and follow professional design standards.
13. **Conviviality** – The interface should be friendly, polite, and supportive.

17 Teikning Senu 10.3 - Essay - Liður 1

1. Draw the scene using JavaFX UI components.
Answer this part on a sketch sheet or, if you prefer, in the drawing section of the editor (pen).

Fill in your answer here

Format

Words: 0

Maximum marks: 5

18 Teikning Senu Liður 2

2. Select UI components that are appropriate for each action (e.g., TextField, ComboBox, DatePicker, Button, or other). Set it up in a table (see the table icon in the editor) with two columns: Action; JavaFX UI Class.

Fill in your answer here

Format

Words: 0

Maximum marks: 5

19 Teikning Senu Liður 3

3. Justify the design with reference to at least two items from the checklist. Mention the items you used with the number and description.

Set it up in a table (see the table icon in the editor) with two columns: Item and checklist number; How the item was used in the scene.

Fill in your answer here

Format

Words: 0

Maximum marks: 5

The skeletons of the two classes are above the Java editor

Program the `CurrencyConverter` processing class and create the constructor and methods `convertToISK(double amount, String currency)` which converts currency to ISK and `getCurrencies()` which retrieves all foreign currencies (USD, EUR).

Use fixed exchange rates (e.g., 1 USD = 127.61 ISK, 1 EUR = 144.91 ISK). Record this in the `CurrencyConverter` processing class. See the hint below.

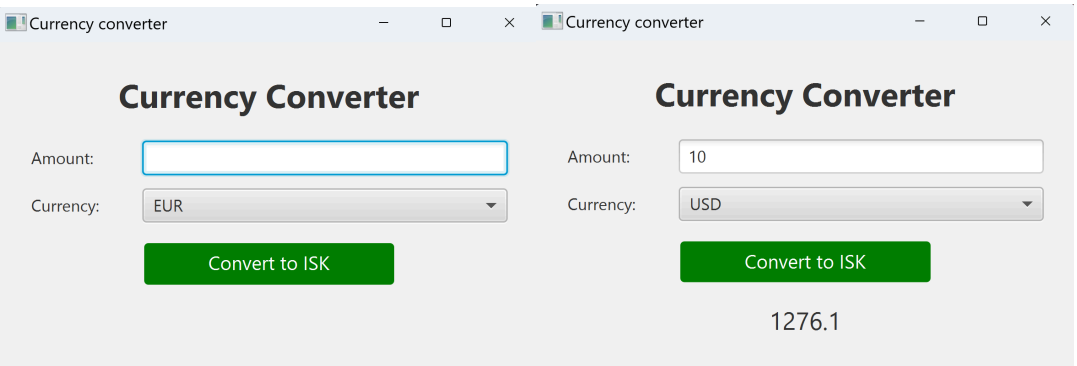
In the `initialize()` method of the `CurrencyController`:

- Create a `CurrencyConverter` object
- Retrieve all currencies from the processor and set them in the `ChoiceBox` (fxCurrency)
- Set the first currency as the current currency in the GUI

In the action handler in `CurrencyController`:

- Read the amount in the foreign currency from the `TextField` (fxText)
- Read the selected currency from the `ChoiceBox` (fxCurrency)
- Call `convertToISK` and display the result in the `Label` (fxResult)

Below is the initial state of the GUI and the state after the user has entered 10, and the program has converted 10 USD to ISK.



Hints

To store the currencies in the back-end, use data structures you are familiar with. Here are two options, but feel free to use the method you prefer or another method.

private final String [] currenciesArray = {"USD", "EUR"};
private final HashMap<String, Double> currencies;

To get ObservableArrayList<String> if you use an array
ObservableArrayList<String> obs = FXCollections.observableArrayList (currenciesArrays);

if you use a HashMap
ObservableArrayList<String> obs = FXCollection.observableArrayList(currencies.keySet())

Look at the following resources that might be useful:
#3 JavaFX API - e.g., FXCollections, `TextField`, `Label`, `ChoiceBox`, `Button`
#6 - Java Cheatsheet for this exam, HashMap, ObservableList
#7 - currency-view.fxml for reference, not required

20 CurrencyConverter forritun - 2.2

```
public class CurrencyController {
    private CurrencyConverter converter;
    @FXML private TextField fxAmount;
    @FXML private ChoiceBox<String> fxCurrency;
    @FXML private Label fxResult;
    public void initialize() {
        // TODO: Create instance of CurrencyConverter

        // TODO: initialize ChoiceBox with currency elements from CurrencyConverter
        // TODO: make the first currency the current one in the GUI

    }
    @FXML
    private void handleConvert(ActionEvent event) {
        // TODO:  read input, call conversion, and display result

    }
}

public class CurrencyConverter {
    // TODO define an array, a list or a hashmap for the currencies of USD and EUR
    public CurrencyConverter() {
        // TODO initialize the chosen data structures for currencies and optionally their values
    }
    public ObservableList<String> getCurrencies() {
        // TODO return an observable list of currency names, i.e. USD and EUR
    }
    public double convertToISK(double amount, String currency) {
        // TODO: Implement conversion logic
    }
}
```

Program in Java

```
1
```

Maximum marks: 15

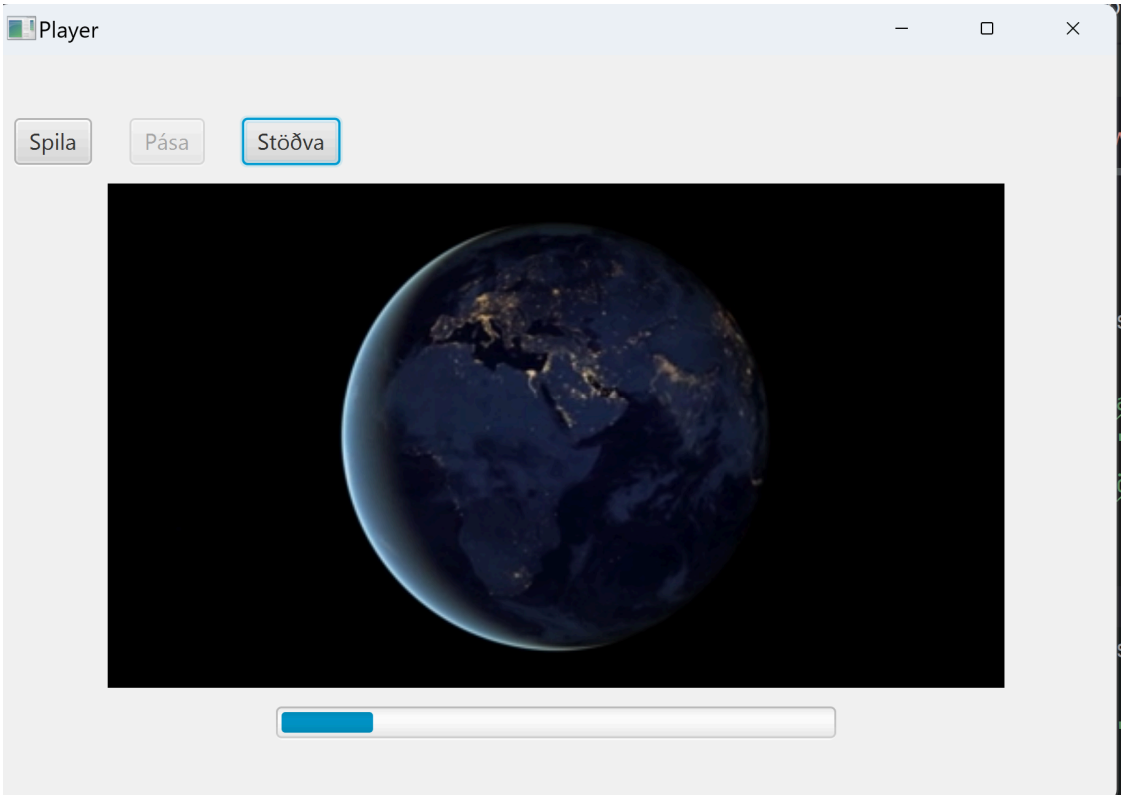
Skeletons for the classes are above the java editor

You are to program the MediaPlayerController that:

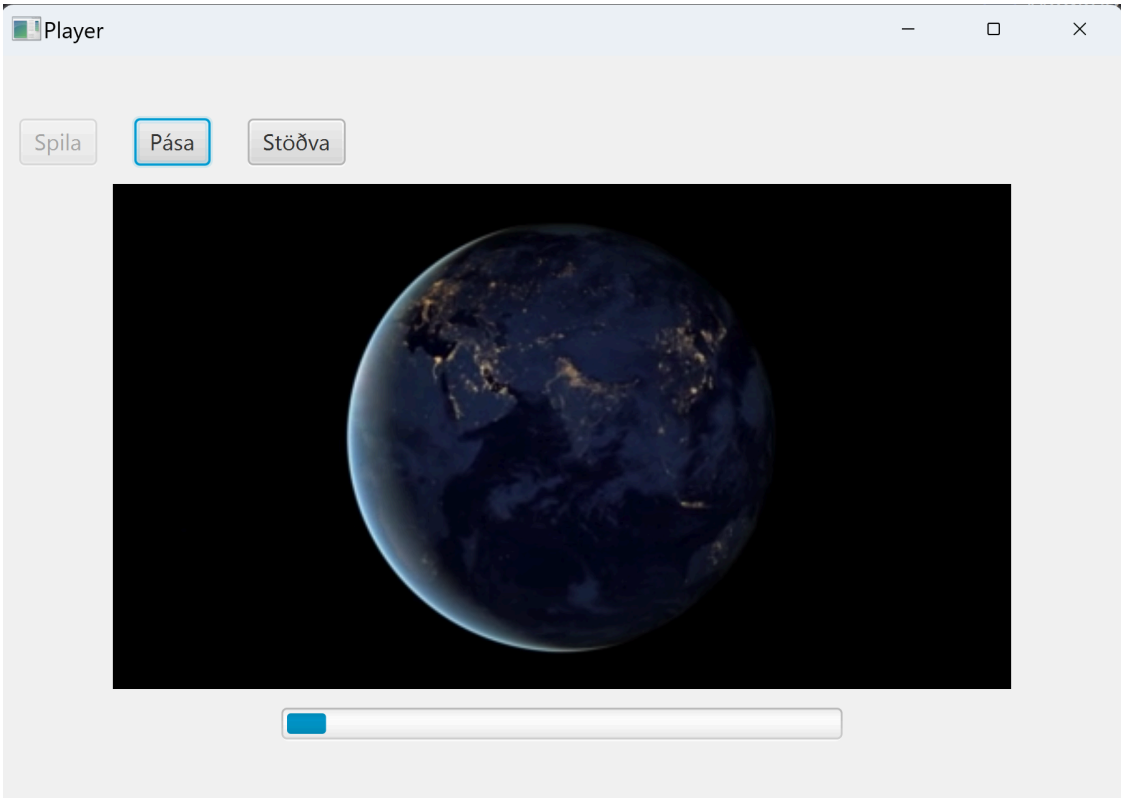
- Creates a `Media` object from a video (.jpeg) file at a fixed path (constant in the code provided).
- Creates a `MediaPlayer` that plays, pauses, and stops the video.

- The play button is disabled when the video is playing, and the pause button is disabled if the video is not playing.
- Displays the video in a `MediaView` UI component.
- Displays the playback progress in a `ProgressBar` that is linked to `currentTimeProperty()` of a MediaPlayer with a `ChangeListener` (using `addListener`).

The player paused



The player playing



The following **resources** may be useful

3 Media, MediaView, MediaPlayer, ProgressBar, Duration, MediaPlayer.Status
#6 Cheatsheet for this exam
#8 media-view.fxml - - for reference, but not necessary to review

21 MediaPlayer 6.3 - forritun

```
public class MediaController {
    private static String MYFILE;
    private static final String MYNAME = "file_example_MP4_480_1_5MG.mp4";
    // GUI variables
    @FXML
    private Button fxSpilaButton, fxPausaButton, fxStoppaButton;
    @FXML
    private MediaView fxMediaView;
    @FXML
    private ProgressBar fxProgressBar;
    private MediaPlayer mediaPlayer;
    @FXML
    private void initialize() {
        MYFILE = String.valueOf(getClass().getResource(MYNAME));
        // TODO: Create Media and MediaPlayer with a fixed file with path in MYFILE
        // TODO: Set up MediaView with MediaPlayer
        // TODO: set up a rule, e.g. binding, making spila button inactive when player is playing
        // TODO: set a a rule, e.g. binding, making pause button inactive when player is not playing
        // TODO: Set up a listener from MediaPlayer's currentTimeProperty to fxProgressBar
       }
    @FXML
    private void spila() {
        // TODO: Start playback
    }
    @FXML
    private void pausa() {
        // TODO: Pause playback
    }
    @FXML
    private void stoppa() {
        // TODO: Stop playback
    }
}
```

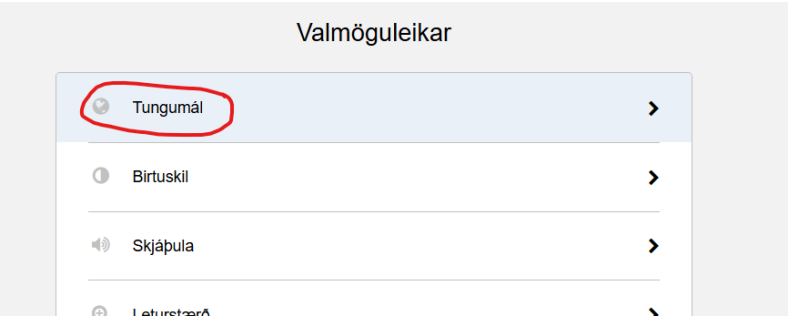
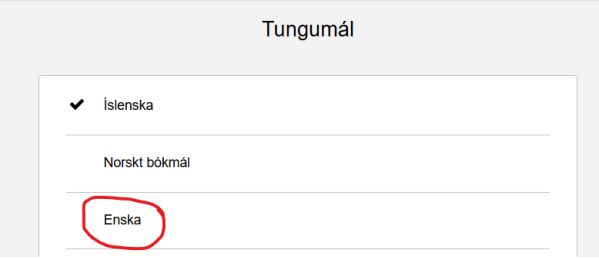
Program in Java

Maximum marks: 15

Document 1
Attached

This exam is offered in Icelandic and English. To change to English please click on the ‚hamburger‘ menu in the top right corner of the exam:

Then click on ‚tungumál‘ or ‚languages‘:



And select ‚Enska‘ or ‚English‘: